Additional Table 5 Characteristics of studies on trehalose intervention for animal models of amyotrophic lateral sclerosis

Studies, country	Blinding of assessor	Intervention(s), Control and setting (Intervention schedule)	Animal characteristics (Neurodegenerative diseases)	Outcome measure(s)	Clinical outcome(s)	Physiological changes	Adverse event(s)	Conclusion
Castillo et al. (2013) [28] Chile	N/R	**Intervention[1]:** Trehalose. (2 g/kg, injection, 3x/wk) **Control[1]:** Glucose/Sucrose (2 g/kg, injection, 3x/wk) **Control[3]:** PBS (2 g/kg, injection,	**SOD1^{G86R} transgenic mice [ALS]** Intervention: n = 17 Control[1]: n = 15 Control[2]: n = 15 **Background** C57BL76 **Age (mon)** 1.17	<u>Clinical</u> Lifespan Disease onset and progression - Rota-rod test - Body weight loss - Visual observation of disease signs <u>Biochemical</u> Level of plague	Trehalose prolonged lifespan in transgenic mice; greater effect was observed in male. *Trehalose delayed disease onset and progression (not sig.).	Trehalose reduced overall levels of SOD1 oligomers and monomers in spinal cord and frontal cortex in transgenic mice. Trehalose reduced GFAP^{+} in transgenic mice. Trehalose protected	N/R	Direct correlations between protective effects of trehalose in ALS, upregulation of autophagy, and decrease in levels of mutant SOD1 aggregation; as well as inhibition of reactive astrogliosis.

	3x/wk)	**Gender** Intervention: 9 Male 8 Female - Disease progression: 7 Male 5 Female Control[1]: 7 Male 8 Female - Disease progression: 6 Male 5 Female Control[2]: 8 Male 7 Female - Disease progression: 4 Male 3 Female	- SOD1 (oligomer, monomer, mRNA) Expression of astrogliosis - $GFAP^+$ Motor neuron survival - $NeuN^+$ expression of autophagy marker - LC3 (molecular and mNRA) - SQSTM1 (molecular and mRNA) - Phospho-MTOR - P70S6 kinase 2 - Other mRNAs (Bcen1, Atg5, Atg7) - Stress responsive transcription factors (FOXO1 and ATF4)		motor neurons in transgenic mice. Trehalose increased level of LC3 and LC3II in transgenic mice. Trehalose reduced level of SQSTM1 in transgenic mice. Trehalose increased expression of autophagy genes in spinal cord and brain cortex of transgenic mice. Trehalose increased expression of nuclear translocation of FOXO1 in spinal cord of transgenic mice.		
Li et al. (2015) [29] China	Yes	**Intervention[1]:** Trehalose. [2%, oral administration, daily, 1 mon] **Intervention[2]:** Trehalose. [2%, oral administration, daily, 2 mon]	**Experiment 1 (90 d)** **$SOD1^{G93A}$ transgenic mice [ALS]** Intervention[1]: *n* = 10 Control: *n* = 10 **WT** Intervention[1]: *n* = 10	**Experiment 1** Biochemical Expression of autophagy marker - LC3 - P62 Level of plagues - SOD1 Expression of astrogliosis - $GFAP^+$ Expression of	**Experiment 2** Trehalose delayed disease onset and progression in transgenic mice. Trehalose prolonged lifespan in transgenic mice. Trehalose improved motor functions in	**Experiment 1** Trehalose increased LC3II in spinal cord of transgenic and WT mice. Trehalose decreased p62 and SOD1 aggregation in transgenic mice. Trehalose reduced	N/R	Trehalose may be a useful add-on therapy in conjunction with other ALS treatment options to alleviate symptoms in the early-stage.

Control: Vehicle. [Drinking water]	Control: *n* = 10 **Background** B6SJL **Age (mon)** 2 **Gender** 40 Female **<u>Experiment 2 (end-stage)</u>** **SOD1^{G93A} transgenic mice [ALS]** Intervention[2] (Lifespan): *n* = 15; (Biochemical): *n* = 5 Control[2] (Lifespan): *n* = 15; (Biochemical): *n* = 5 **WT** Intervention[2] (Lifespan): *n* = 10; (Biochemical): *n* = 5 Control[2] (Lifespan): *n* = 10; (Biochemical): *n* = 5 **Background** B6SJL **Age (mon)** 2 **Gender** 50 Female	microgliosis - CD11b **<u>Experiment 2</u>** <u>Clinical</u> Disease onset and progression - Neurological score scale Rota-rod test - Motor coordination Hanging wire test - Motor function <u>Biochemical</u> Expression of autophagy marker - LC3 - P62 Level of plagues - SOD1 Motor neuron survival Expression of astrogliosis - GFAP^{+} Expression of microgliosis - CD11b	transgenic mice in the early stage; effect disappeared in the later stage.	GFAP^{+} and CD11b levels in transgenic mice. **<u>Experiment 2</u>** Trehalose increased LC3II in transgenic mice. Protective effect of motor neuron in lumbar spinal cords of transgenic mice. Trehalose reduced GFAP^{+} and CD11b levels in transgenic mice.

Zhang et al. (2014) [30] Yes China	**Intervention:** Trehalose. [2%, oral administration, daily, until death] **Control[1]:** [2%, oral administration, daily, until death] **Control[2]:** Vehicle. [None]	**SOD1^{G93A} transgenic mice [ALS]** Intervention: n = 21 Control[1]: n = 21 Control[2]: n = 21 **WT** Control[2]: n = 15 **Background** B6SJL **Age (mon)** 2.13 **Gender** 78 Male	<u>Clinical</u> Disease onset and progression - Rota-rod test (Motor coordination) Lifespan <u>Biochemical</u> Motor neuron survival - Number - SMI-32^{+} Skeletal muscle denervation - HE^{+} - NADH^{+} Oxidative stress - MDA NMJ degeneration - NSE^{+} - ACHE^{+} Expression of autophagy - LC3II - ATG5 - MTOR (p-MTOR, p-RPS6KB, p-AKT1) Level of plague - SOD1 (molecular and mRNA) - SQSTM1 - ubiquitin^{+} Mitochondria degeneration - number - diameter Ultra-structure alterations	Trehalose delayed disease onset in transgenic mice. Trehalose prolonged lifespan in transgenic mice.	Trehalose increased number of motor neurons and SMI-32^{+} in transgenic mice. Trehalose attenuated muscle degeneration (HE^{+}) and increased the fibre area of transgenic mice. Trehalose increased oxidative metabolism (NADH^{+}) and type I or type II myofibres in transgenic mice. Trehalose decreased MDA in skeletal muscle of transgenic mice. Trehalose increased the number of NMJ in gastrocnemius muscle of transgenic mice. Trehalose reduced SQSTM1 and SOD1 aggregates and in motor neurons of transgenic mice. Trehalose reduced ubiquitin^{+} aggregates in spinal cord of transgenic mice. Trehalose reduced diameter of mitochondria in transgenic mice.	N/R	Trehalose is neuroprotective in ALS and autophagosome-lysosome fusion is a possible therapeutic target.

	- number of autophagosomes and autolysosomes Apoptotic process - BAX - Cytochrome c - Cleaved caspase-3 - Cleaved PARP1	Trehalose improved autolysosome/ autophagosome ratio in motor neurons of transgenic mice. Trehalose decreased levels of BAX, cytochrome c, cleaved PARP1 and caspase-3 in transgenic mice.	

[+]: Positive; ACHE: acetylcholinesterase; ALS: amyotrophic lateral sclerosis; ATF4: activating transcription factor 4; ATG5: autophagy related 5; BAX: Bcl-2-associated X protein; FOXO1: Forkhead Box O1; GFAP: glial fibrillary acidic protein; HE: hematoxylin-eosin; LC3: microtubule-associated protein 1A/1B-light chain 3; MDA: malondialdehyde; MTOR: mammalian target of rapamycin; N/R: not reported; NADH: nicotinamide adenine dinucleotide + hydrogen; NSE: neuron specific enolase; PARP1: poly [ADP-ribose] polymerase 1; RPS6KB: ribosomal protein S6 kinase beta-1; SMI-32: anti-neurofilament H non-phosphorylated mouse mAb; SOD1: superoxide dismutase 1; SQSTM1: sequestosome 1; WT: wild-type

1. Du J, Liang Y, Xu F, Sun B, Wang Z. Trehalose rescues Alzheimer's disease phenotypes in APP/PS1 transgenic mice. J Pharm Pharmacol 2013;65:1753-1756.
2. Lee YS, Lai DM, Huang HJ, Lee-Chen GJ, Chang CH, Hsieh-Li HM, Lee GC. Prebiotic lactulose ameliorates the cognitive deficit in alzheimer's disease mouse model through macroautophagy and chaperone-mediated autophagy pathways. J Agric Food Chem 2021;69:2422-2437.
3. Liu Y, Wang J, Hsiung GR, Song W. Trehalose inhibits Aβ generation and plaque formation in Alzheimer's disease. Mol Neurobiol 2020;57:3150-3157.
4. Perucho J, Casarejos MJ, Gomez A, Solano RM, de Yébenes JG, Mena MA. Trehalose protects from aggravation of amyloid pathology induced by isoflurane anesthesia in APP(swe) mutant mice. Curr Alzheimer Res 2012;9:334-343.
5. Perucho J, Casarejos MJ, Gomez A, Solano RM, de Yébenes JG, Mena MA. Trehalose protects from aggravation of amyloid pathology induced by isoflurane anesthesia in APP(swe) mutant mice. Curr Alzheimer Res 2012;9:334-343.
6. Pupyshev AB, Belichenko VM, Tenditnik MV, Bashirzade AA, Dubrovina NI, Ovsyukova MV, Akopyan AA, Fedoseeva LA, Korolenko TA, Amstislavskaya TG, Tikhonova MA. Combined induction of mTOR-dependent and mTOR-independent pathways of autophagy activation as an experimental therapy for Alzheimer's disease-like pathology in a mouse model. Pharmacol Biochem Behav. 2022;217:173406.
7. Holler CJ, Taylor G, McEachin ZT, Deng Q, Watkins WJ, Hudson K, Easley CA, Hu WT, Hales CM, Rossoll W, Bassell GJ, Kukar T. Trehalose upregulates progranulin expression in human and mouse models of GRN haploinsufficiency: a novel therapeutic lead to treat frontotemporal dementia. Mol Neurodegener 2016;11:46.
8. Rodríguez-Navarro JA, Rodríguez L, Casarejos MJ, Solano RM, Gómez A, Perucho J, Cuervo AM, García de Yébenes J, Mena MA. Trehalose ameliorates dopaminergic and tau pathology in parkin deleted/tau overexpressing mice through autophagy activation. Neurobiol Dis 2010;39):423-438.
9. Schaeffer V, Lavenir I, Ozcelik S, Tolnay M, Winkler DT, Goedert M. Stimulation of autophagy reduces neurodegeneration in a mouse model of human tauopathy. Brain 2012;135(Pt 7):2169-2177.
10. Schaeffer V, Goedert M. Stimulation of autophagy is neuroprotective in a mouse model of human tauopathy. Autophagy 2012;8:1686-1687.
11. Darabi S, Noori-Zadeh A, Abbaszadeh HA, Rajaei F, Bakhtiyari S. Trehalose neuroprotective effects on the substantia nigra dopaminergic cells by activating autophagy and non-canonical Nrf2 pathways. Iran J Pharm Res 2019;18:1419-1428.
12. Ferguson SA, Law CD, Sarkar S. Chronic MPTP treatment produces hyperactivity in male mice which is not alleviated by concurrent trehalose treatment. Behav Brain Res 2015;292:68-78.

13. He Q, Koprich JB, Wang Y, Yu WB, Xiao BG, Brotchie JM, Wang J. Treatment with trehalose prevents behavioral and neurochemical deficits produced in an AAV α-synuclein rat model of Parkinson's disease. Mol Neurobiol 2016;53:2258-25268.
14. Howson PA, Johnston TH, Ravenscroft P, Hill MP, Su J, Brotchie JM, Koprich JB. Beneficial effects of trehalose on striatal dopaminergic deficits in rodent and primate models of synucleinopathy in Parkinson's disease. J Pharmacol Exp Ther. 2019;369:364-374.
15. Kakoty V, K C S, Dubey SK, Yang CH, Taliyan R. Neuroprotective effects of trehalose and sodium butyrate on preformed fibrillar form of α-synuclein-induced rat model of Parkinson's disease. ACS Chem Neurosci. 2021;12:2643-2660.
16. Lin CH, Wei PC, Chen CM, Huang YT, Lin JL, Lo YS, Lin JL, Lin CY, Wu YR, Chang KH, Lee-Chen GJ. Lactulose and melibiose attenuate MPTP-induced Parkinson's disease in mice by inhibition of oxidative stress, reduction of neuroinflammation and up-regulation of autophagy. Front Aging Neurosci 2020;12:226.
17. Pupyshev AB, Tikhonova MA, Akopyan AA, Tenditnik MV, Dubrovina NI, Korolenko TA. Therapeutic activation of autophagy by combined treatment with rapamycin and trehalose in a mouse MPTP-induced model of Parkinson's disease. Pharmacol Biochem Behav 2019;177:1-11.
18. Pupyshev AB, Tenditnik MV, Ovsyukova MV, Akopyan AA, Dubrovina NI, Tikhonova MA. Restoration of Parkinson's disease-like deficits by activating autophagy through mTOR-dependent and mTOR-independent mechanisms in pharmacological and transgenic models of Parkinson's disease in mice. Bull Exp Biol Med. 2021;171:425-430.
19. Sarkar S, Chigurupati S, Raymick J, Mann D, Bowyer JF, Schmitt T, Beger RD, Hanig JP, Schmued LC, Paule MG. Neuroprotective effect of the chemical chaperone, trehalose in a chronic MPTP-induced Parkinson's disease mouse model. Neurotoxicology 2014;44:250-262.
20. Tanji K, Miki Y, Maruyama A, Mimura J, Matsumiya T, Mori F, Imaizumi T, Itoh K, Wakabayashi K. Trehalose intake induces chaperone molecules along with autophagy in a mouse model of Lewy body disease. Biochem Biophys Res Commun 2015;465:746-752.
21. Debnath K, Pradhan N, Singh BK, Jana NR, Jana NR. Poly(trehalose) nanoparticles prevent amyloid aggregation and suppress polyglutamine aggregation in a Huntington's disease model mouse. ACS Appl Mater Interfaces 2017;9:24126-24139.
22. Debnath K, Pradhan N, Singh BK, Jana NR, Jana NR. Correction to poly(trehalose) nanoparticles prevent amyloid aggregation and suppress polyglutamine aggregation in a Huntington's disease model mouse. ACS Appl Mater Interfaces. 2019;11:30508.
23. Im JK, Kim SJ, Jeong YH, Kim WI, Lee DH, Lee WS, Chang YT, Kim KT, Chung SK. Preparation and evaluation of BBB-permeable trehalose derivatives as potential therapeutic agents for Huntington's disease. Med Chem Comm 2013; 4: 310-316.
24. Tanaka M, Machida Y, Niu S, Ikeda T, Jana NR, Doi H, Kurosawa M, Nekooki M, Nukina N. Trehalose alleviates polyglutamine-mediated pathology in a mouse model of Huntington disease. Nat Med 2004;10:148-154.
25. Yang CR, Yu RK. Intracerebral transplantation of neural stem cells combined with trehalose ingestion alleviates pathology in a mouse model of Huntington's disease. J Neurosci Res 2009;87:26-33.

26. Santana MM, Paixão S, Cunha-Santos J, Silva TP, Trevino-Garcia A, Gaspar LS, Nóbrega C, Nobre RJ, Cavadas C, Greif H, Pereira de Almeida L. Trehalose alleviates the phenotype of Machado-Joseph disease mouse models. J Transl Med. 2020;18:161.
27. Chen ZZ, Wang CM, Lee GC, Hsu HC, Wu TL, Lin CW, Ma CK, Lee-Chen GJ, Huang HJ, Hsieh-Li HM. Trehalose attenuates the gait ataxia and gliosis of spinocerebellar ataxia type 17 mice. Neurochem Res 2015;40(4):800-810.
28. Castillo K, Nassif M, Valenzuela V, Rojas F, Matus S, Mercado G, Court FA, van Zundert B, Hetz C. Trehalose delays the progression of amyotrophic lateral sclerosis by enhancing autophagy in motoneurons. Autophagy 2013;9:1308-1320.
29. Li Y, Guo Y, Wang X, Yu X, Duan W, Hong K, Wang J, Han H, Li C. Trehalose decreases mutant SOD1 expression and alleviates motor deficiency in early but not end-stage amyotrophic lateral sclerosis in a SOD1-G93A mouse model. Neuroscience 2015;298:12-25.
30. Zhang X, Chen S, Song L, Tang Y, Shen Y, Jia L, Le W. MTOR-independent, autophagic enhancer trehalose prolongs motor neuron survival and ameliorates the autophagic flux defect in a mouse model of amyotrophic lateral sclerosis. Autophagy 2014;10:588-602.s